

BREATHEBOX · DESIGN DECISIONS REGISTER

BreatheBox design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from REVIEW.md, BBX-DDR-002 and BBX-DDR-003

BreatheBox design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Open decisions, all Proposed, awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Design for construction: accept the changes that make the concept buildable (housing boards and battens, core frames, dividers and fan bulkhead, filter seats, grilles, bolted window insert, rigid rail-and-strut bracket with the foot at 500 mm, drain and power entry)	Accept as made; or ask for changes item by item	Accept	The whole build plan	BBX-DDR-003, P1 to P12
2	Sliding restraint: only sill friction (needed 0.40, about 0.6 available on anti-slip tape) stops the unit sliding into the room and tipping off the sill	(a) accept, with a 50 N push test at TRL 4; (b) lower the foot to 400 mm (margin about 1.9, longer struts); (c) a positive restraint such as a strap or a clamp to the window board, which may need drilling	(a), moving to (b) if the push test fails	Wall foot height and strut length (sections 3.12 and 3.13); first checks	BBX-DDR-003, A2
3	Sash lock (R12, safety case): no part locks the raised sash onto the insert panel, and the window's own catch no longer meets	(a) a bought no-drill sash jammer or adjustable bar in the inner track above the raised sash, about \$10; (b) screw-fixed sash stops, which need drilling (R9)	(a)	Window fitting (step 15); bill of materials	BBX-DDR-003, A3
4	Fan position: both fans at the room end put the exhaust side of the core at the higher pressure, so any core leakage reaches the supply	Keep both fans at the room end (warm, dry, easy to reach); or move the exhaust fan to the outdoor end (cold, wet air)	None made	Fan bulkhead and fan positions (section 3.6)	BBX-DDR-002, O4
5	First co-design partner and region (for example a tenants' group or social landlord in the UK or Canada)	Open	None made	Trial window and its sill, stop bead and wall	BBX-DDR-001 and BBX-DDR-002, O1
6	Status light and button on the room face, in a small pod (appearance model)	Adopt the room-face position; or keep them inside on the controller board	Adopt; the build plan already puts them behind two holes in the room face	Room end board holes and status board (sections 3.1 and 3.8)	REVIEW.md, 2026-09-26, item 2
7	Clear inspection window in the lid (appearance model)	Keep for the renders and decide at TRL 4 on a clear lid's cost and fire rating; or an opaque lid	Keep for the renders, decide at TRL 4; the build plan uses an opaque lid	Lid (section 3.11)	REVIEW.md, 2026-09-26, item 1

#	Decision needed	Options	Recommendation	Affects in the build	Source
8	Lid joint: the appearance model splits the lid 40 mm below the top with side latches; the constructable design's lid is the top board alone, with two toggle latches	Top board lid as built; or the appearance model's deeper lid	Top board lid; update the appearance model to match	Lid and latches (section 3.11)	REVIEW.md, 2026-09-26, item 3; BBX-DDR-003, P1

■ To confirm when parts are bought

Table 2. Items to confirm when parts are bought or the trial window is chosen.

#	What to confirm	Why it matters	Source
1	The counterflow core's ends are each split into two halves along a centre line, one per air stream, and it is 180 x 180 x 300 mm	The core frames' U and the dividers' tongues are made for that port layout; a hexagonal core with side ports needs different frames	BBX-DDR-003, P2
2	The blowers' inlet ring is 80 mm across or less, their mounting holes, and that the supply fan's outlet can face up toward the lid grille	Sets the bulkhead holes and fan mounting; the calculations use class values for the fan curve and noise	BBX-DDR-003, P3; BBX-CAL-001 section 11
3	The supply filter is 150 x 170 x 25 mm with a frame at least 5 mm wide; the exhaust pad is 220 x 190 x 12 mm	The lining lips seal on the filter frame and the pad edge	BBX-DDR-003, P5
4	The trial window's sill is at least 110 mm deep, its stop bead fits the 17 mm gap between the housing and the collar flange, and the wall below is clear for the foot 470 to 530 mm above the floor (no radiator)	Sets the collar length and whether the bracket fits as drawn	BBX-DDR-003, P8 and P9
5	The anti-slip tape grips the trial sill's finish with friction of about 0.6 or more	The unit needs 0.40 to stay put	BBX-DDR-003, A2
6	The toggle latches fit 6 mm board, and the latch screws can be backed by nuts inside	Lid fixing	BBX-DDR-003, P1

■ Value engineering

Value-engineering target: USD 255 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 283 (USD 28 over the target). Main cost drivers and savings worth trying:

- The largest lines are the counterflow core (USD 48), the insulated housing (USD 42), the controller with CO₂ and humidity sensing (USD 42), the outdoor hoods and collars (USD 26), the window insert panel (USD 20) and the two fans (USD 19 each).
- The concept's bill of materials was USD 255. The rise came from parts the concept needed but did not list (BBX-DDR-003): partitions, filter seats, grilles and latches (USD 18), hood back plates and collar flanges (USD 2), the bracket's rails, cleats and foot (USD 4), and bolts, gland and anti-slip tape (USD 4).
- Savings worth trying: printed grilles and a cheaper core, at some cost in performance. A bought sash jammer (about USD 10, open decision 3) would add to the estimate.

■ Decisions made

Table 3. Decisions made by Amish.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 items A1 to A6: budget raised, sash windows first, bought polymer counterflow core, frost by slowing the supply fan, 50 m³/h nominal with boost to 70 m³/h, ESP32-C3 class controller with an SCD41 class sensor	Amish: "i accept all your recommendations, go with them across all repos."	BBX-DDR-001, BBX-DDR-002
2026-09-25	Noise (R6): quiet night mode at 32 m³/h, plus larger slower fans or a lined silencer checked against a real datasheet before any change to R6 (on hold with TRL 4)	Amish, same instruction	BBX-DDR-002, item 7
2026-09-25	Larger blowers (about 100 m³/h free air, 400 Pa); 2 A time-delay input fuse; firmware fan speed cap; no use in a room whose door seals airtight	Amish, same instruction	BBX-DDR-002, items 8 to 11
2026-09-25	Check the CO2 sensor on the lab's CalRig before use (on hold with TRL 4)	Amish, same instruction	BBX-DDR-002, item 6
2026-09-26	Budget set to \$255 to cover the concept's priced BOM	Amish: "i approve all the budget items."	BBX-DDR-002 v0.2
2026-09-30	Build plans in the approved format, with outstanding decisions kept in this separate register, not in the build plan	Amish: "this is the correct build plan ... Extend this across all the other repos"; "don't log outstanding decisions in this build plan"	BBX-BLD-001; this register
2026-09-30	Where the concept cannot be built, fix the design assumptions so it is physically feasible (instruction; the resulting changes are open item 1 above)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	BBX-DDR-003